

VI.

The Relation of Quantum Theory to Other Parts of Natural Science

IT HAS been stated before that the concepts of natural science can sometimes be sharply defined with regard to their connections. This possibility was realized for the first time in Newton's *Principia* and it is just for that reason that Newton's work has exerted its enormous influence on the whole development of natural science in the following centuries. Newton begins his *Principia* with a group of definitions and axioms which are interconnected in such a way that they form what one may call a "closed system." Each concept can be represented by a mathematical symbol, and the connections between the different concepts are then represented by mathematical equations expressed by means of the symbols. The mathematical image of the system ensures that contradictions cannot occur in the system. In this way the possible motions of bodies under the influence of the acting forces are represented by the possible solutions of the equations. The system of definitions and axioms which can be written in a set of mathematical equations is considered as describing an eternal structure of nature, depending neither on a particular space nor on particular time.

The connection between the different concepts in the system is so close that one could generally not change any one of the concepts without destroying the whole system.

For this reason Newton's system was for a long time considered as final and the task set before the scientists of the following period seemed simply to be an expansion of Newton's mechanics into wider fields of experience. Actually physics did develop along these lines for about two centuries.

From the theory of the motion of mass points one could go over to the mechanics of solid bodies, to rotatory motions, and one could treat the continuous motions of a fluid or the vibrating motions of an elastic body. All these parts of mechanics or dynamics were gradually developed in close connection with the evolution of mathematics, especially of the differential calculus, and the results were checked by experiments. Acoustics and hydrodynamics became a part of mechanics. Another science, in which the application of Newton's mechanics was obvious, was astronomy. The improvements of the mathematical methods gradually led to more and more accurate determinations of the motions of the planets and of their mutual interactions. When the phenomena of electricity and magnetism were discovered, the electric or magnetic forces were compared to the gravitational forces and their actions upon the motion of the bodies could again be studied along the lines of Newtonian mechanics. Finally, in the nineteenth century, even the theory of heat could be reduced to mechanics by the assumption that heat really consists of a complicated statistical motion of the smallest parts of matter. By combining the concepts of the mathematical theory of probability with the concepts of Newtonian mechanics Clausius, Gibbs and Boltzmann were able to show that the fundamental laws in the theory of heat could be interpreted as

statistical laws following from Newton's mechanics when applied to very complicated mechanical systems.

Up to this point the program set up by Newtonian mechanics had been carried out quite consistently and had led to the understanding of a wide field of experience. The first difficulty arose in the discussions on the electromagnetic field in the work of Faraday and Maxwell. In Newtonian mechanics the gravitational force had been considered as given, not as an object for further theoretical studies. In the work of Faraday and Maxwell, however, the field of force itself became the object of the investigation; the physicists wanted to know how this field of force varied as function of space and time. Therefore, they tried to set up equations of motion for the fields, not primarily for the bodies upon which the fields act. This change led back to a point of view which had been held by many scientists before Newton. An action could, so it seemed, be transferred from one body to another only when the two bodies touched each other; for instance, in a collision or through friction. Newton had introduced a very new and strange hypothesis by assuming a force that acted over a long distance. Now in the theory of the fields of force one could come back to the older idea, that action is transferred from one point to a neighboring point, only by describing the behavior of the fields in terms of differential equations. This proved actually to be possible, and therefore the description of the electromagnetic fields as given by Maxwell's equations seemed a satisfactory solution of the problem of force. Here one had really changed the program given by Newtonian mechanics. The axioms and definitions of Newton had referred to bodies and their motion; but with Maxwell the fields of force seemed to have acquired the same degree of reality as the bodies in Newton's theory. This view of course was not easily ac-

cepted; and to avoid such a change in the concept of reality it seemed plausible to compare the electromagnetic fields with the fields of elastic deformation or stress, the light waves of Maxwell's theory with the sound waves in elastic bodies. Therefore, many physicists believed that Maxwell's equations actually referred to the deformations of an elastic medium, which they called the ether; and this name was given merely to explain that the medium was so light and thin that it could penetrate into other matter and could not be seen or felt. This explanation was not too satisfactory, however, since it could not explain the complete absence of any longitudinal light waves.

Finally the theory of relativity, which will be discussed in the next chapter, showed in a conclusive way that the concept of the ether as a substance, to which Maxwell's equations refer, had to be abandoned. The arguments cannot be discussed at this point; but the result was that the fields had to be considered as an independent reality.

A further and still more startling result of the theory of special relativity was the discovery of new properties of space and time, actually of a relation between space and time that had not been known before and did not exist in Newtonian mechanics.

Under the impression of this completely new situation many physicists came to the following somewhat rash conclusion: Newtonian mechanics had finally been disproved. The primary reality is the field and not the body, and the structure of space and time is correctly described by the formulas of Lorentz and Einstein, and not by the axioms of Newton. The mechanics of Newton was a good approximation in many cases, but now it must be improved to give a more rigorous description of nature.

From the point of view which we have finally reached in

quantum theory such a statement would appear as a very poor description of the actual situation. First, it ignores the fact that most experiments by which fields are measured are based upon Newtonian mechanics and, second, that Newtonian mechanics cannot be improved; it can only be replaced by something essentially different!

The development of quantum theory has taught us that one should rather describe the situation in the following terms: Wherever the concepts of Newtonian mechanics can be used to describe events in nature, the laws formulated by Newton are strictly correct and cannot be improved. But the electromagnetic phenomena cannot adequately be described by the concepts of Newtonian mechanics. Therefore, the experiments on the electromagnetic fields and on light waves, together with their theoretical analysis by Maxwell, Lorentz and Einstein, have led to a new closed system of definitions and axioms and of concepts that can be represented by mathematical symbols, which is coherent in the same sense as the system of Newton's mechanics, but is essentially different from it.

Therefore, even the hopes which had accompanied the work of the scientists since Newton had to be changed. Apparently progress in science could not always be achieved by using the known laws of nature for explaining new phenomena. In some cases new phenomena that had been observed could only be understood by new concepts which were adapted to the new phenomena in the same way as Newton's concepts were to the mechanical events. These new concepts again could be connected in a closed system and represented by mathematical symbols. But if physics or, more generally, natural science proceeded in this way, the question arose: What is the relation between the different sets of concepts? If, for instance, the same

concepts or words occur in two different sets and are defined differently with regard to their connection and mathematical representation, in what sense do the concepts represent reality?

This problem arose at once when the theory of special relativity had been discovered. The concepts of space and time belonged both to Newtonian mechanics and to the theory of relativity. But space and time in Newtonian mechanics were independent; in the theory of relativity they were connected by the Lorentz transformation. In this special case one could show that the statements of the theory of relativity approached those of Newtonian mechanics within the limit in which all velocities in the system are very small as compared with the velocity of light. From this one could conclude that the concepts of Newtonian mechanics could not be applied to events in which there occurred velocities comparable to the velocity of light. Thereby one had finally found an essential limitation of Newtonian mechanics which could not be seen from the coherent set of concepts nor from simple observations on mechanical systems.

Therefore, the relation between two different coherent sets of concepts always requires very careful investigation. Before we enter into a general discussion about the structure of any such closed and coherent set of concepts and about their possible relations we will give a brief description of those sets of concepts that have so far been defined in physics. One can distinguish four systems that have already attained their final form.

The first set, Newtonian mechanics, has already been discussed. It is suited for the description of all mechanical systems, of the motion of fluids and the elastic vibration of bodies; it comprises acoustics, statics, aerodynamics.

The second closed system of concepts was formed in the course of the nineteenth century in connection with the theory

of heat. Though the theory of heat could finally be connected with mechanics through the development of statistical mechanics, it would not be realistic to consider it as a part of mechanics. Actually the phenomenological theory of heat uses a number of concepts that have no counterpart in other branches of physics, like: heat, specific heat, entropy, free energy, etc. If from this phenomenological description one goes over to a statistical interpretation, by considering heat as energy, distributed statistically among the very many degrees of freedom due to the atomic structure of matter, then heat is no more connected with mechanics than with electrodynamics or other parts of physics. The central concept of this interpretation is the concept of probability, closely connected with the concept of entropy in the phenomenological theory. Besides this concept the statistical theory of heat requires the concept of energy. But any coherent set of axioms and concepts in physics will necessarily contain the concepts of energy, momentum and angular momentum and the law that these quantities must under certain conditions be conserved. This follows if the coherent set is intended to describe certain features of nature that are correct at all times and everywhere; in other words, features that do not depend on space and time or, as the mathematicians put it, that are invariant under arbitrary translations in space and time, rotations in space and the Galileo—or Lorentz—transformation. Therefore, the theory of heat can be combined with any of the other closed systems of concepts.

The third closed system of concepts and axioms has its origin in the phenomena of electricity and magnetism and has reached its final form in the first decade of the twentieth century through the work of Lorentz, Einstein and Minkowski. It comprises electrodynamics, special relativity, optics, magnetism, and one

may include the de Broglie theory of matter waves of all different sorts of elementary particles, but not the wave theory of Schrödinger.

Finally, the fourth coherent system is essentially the quantum theory as it has been described in the first two chapters. Its central concept is the probability function, or the "statistical matrix," as the mathematicians call it. It comprises quantum and wave mechanics, the theory of atomic spectra, chemistry, and the theory of other properties of matter like electric conductivity, ferromagnetism, etc.

The relations between these four sets of concepts can be indicated in the following way: The first set is contained in the third as the limiting case where the velocity of light can be considered as infinitely big, and is contained in the fourth as the limiting case where Planck's constant of action can be considered as infinitely small. The first and partly the third set belong to the fourth as a priori for the description of the experiments. The second set can be connected with any of the other three sets without difficulty and is especially important in its connection with the fourth. The independent existence of the third and fourth sets suggests the existence of a fifth set, of which one, three, and four are limiting cases. This fifth set will probably be found someday in connection with the theory of the elementary particles.

We have omitted from this enumeration the set of concepts connected with the theory of general relativity, since this set has perhaps not yet reached its final form. But it should be emphasized that it is distinctly different from the other four sets.

After this short survey we may come back to the more general question, what one should consider as the characteristic features of such a closed system of axioms and definitions. Perhaps the

most important feature is the possibility of finding a consistent mathematical representation for it. This representation must guarantee that the system does not contain contradictions. Then the system must be suited to describe a wide field of experience. The great variety of phenomena in the field should correspond to the great number of solutions of the equations in the mathematical representation. The limitations of the field can generally not be derived from the concepts. The concepts are not sharply defined in their relation to nature, in spite of the sharp definition of their possible connections. The limitations will therefore be found from experience, from the fact that the concepts do not allow a complete description of the observed phenomena.

After this brief analysis of the structure of present-day physics the relation between physics and other branches of natural science may be discussed. The nearest neighbor to physics is chemistry. Actually through quantum theory these two sciences have come to a complete union. But a hundred years ago they were widely separated, their methods of research were quite different, and the concepts of chemistry had at that time no counterpart in physics. Concepts like valency, activity, solubility and volatility had a more qualitative character, and chemistry scarcely belonged to the exact sciences. When the theory of heat had been developed by the middle of the last century scientists started to apply it to the chemical processes, and ever since then the scientific work in this field has been determined by the hope of reducing the laws of chemistry to the mechanics of the atoms. It should be emphasized, however, that this was not possible within the framework of Newtonian mechanics. In order to give a quantitative description of the laws of chemistry one had to formulate a much wider system of concepts for atomic physics. This was finally done in quantum theory, which has its roots

just as much in chemistry as in atomic physics. Then it was easy to see that the laws of chemistry could not be reduced to Newtonian mechanics of atomic particles, since the chemical elements displayed in their behavior a degree of stability completely lacking in mechanical systems. But it was not until Bohr's theory of the atom in 1913 that this point had been clearly understood. In the final result, one may say, the concepts of chemistry are in part complementary to the mechanical concepts. If we know that an atom is in its lowest stationary state that determines its chemical properties we cannot at the same time speak about the motion of the electrons in the atom.

The present relation between biology, on the one side, and physics and chemistry, on the other, may be very similar to that between chemistry and physics a hundred years ago. The methods of biology are different from those of physics and chemistry, and the typical biological concepts are of a more qualitative character than those of the exact sciences. Concepts like life, organ, cell, function of an organ, perception have no counterpart in physics or chemistry. On the other hand, most of the progress made in biology during the past hundred years has been achieved through the application of chemistry and physics to the living organism, and the whole tendency of biology in our time is to explain biological phenomena on the basis of the known physical and chemical laws. Again the question arises, whether this hope is justified or not.

Just as in the case of chemistry, one learns from simple biological experience that the living organisms display a degree of stability which general complicated structures consisting of many different types of molecules could certainly not have on the basis of the physical and chemical laws alone. Therefore, something

has to be added to the laws of physics and chemistry before the biological phenomena can be completely understood.

With regard to this question two distinctly different views have frequently been discussed in the biological literature. The one view refers to Darwin's theory of evolution in its connection with modern genetics. According to this theory, the only concept which has to be added to those of physics and chemistry in order to understand life is the concept of history. The enormous time interval of roughly four thousand million years that has elapsed since the formation of the earth has given nature the possibility of trying an almost unlimited variety of structures of groups of molecules. Among these structures there have finally been some that could reduplicate themselves by using smaller groups from the surrounding matter, and such structures therefore could be created in great numbers. Accidental changes in the structures provided a still larger variety of the existing structures. Different structures had to compete for the material drawn from the surrounding matter and in this way, through the "survival of the fittest," the evolution of living organisms finally took place. There can be no doubt that this theory contains a very large amount of truth, and many biologists claim that the addition of the concepts of history and evolution to the coherent set of concepts of physics and chemistry will be amply sufficient to account for all biological phenomena. One of the arguments frequently used in favor of this theory emphasizes that wherever the laws of physics and chemistry have been checked in living organisms they have always been found to be correct; there seems definitely to be no place at which some "vital force" different from the forces in physics could enter.

On the other hand, it is just this argument that has lost much of its weight through quantum theory. Since the concepts of

physics and chemistry form a closed and coherent set, namely, that of quantum theory, it is necessary that wherever these concepts can be used to describe phenomena the laws connected with the concepts must be valid too. Therefore, wherever one treats living organisms as physicochemical systems, they must necessarily act as such. The only question from which we can learn something about the adequacy of this first view is whether the physicochemical concepts allow a *complete* description of the organisms. Biologists, who answer this question in the negative, generally hold the second view, that has now to be explained.

This second view can perhaps be stated in the following terms: It is very difficult to see how concepts like perception, function of an organ, affection could be a part of the coherent set of the concepts of quantum theory combined with the concept of history. On the other hand, these concepts are necessary for a complete description of life, even if for the moment we exclude mankind as presenting new problems beyond biology. Therefore, it will probably be necessary for an understanding of life to go beyond quantum theory and to construct a new coherent set of concepts, to which physics and chemistry may belong as "limiting cases;" History may be an essential part of it, and concepts like perception, adaptation, affection also will belong to it. If this view is correct, the combination of Darwin's theory with physics and chemistry would not be sufficient to explain organic life; but still it would be true that living organisms can to a large extent be considered as physicochemical systems—as machines, as Descartes and Laplace have put it—and would, if treated as such, also react as such. One could at the same time assume, as Bohr has suggested, that our knowledge of a cell being alive may be complementary to the complete knowledge of its molecular structure. Since a complete knowledge of

this structure could possibly be achieved only by operations that destroy the life of the cell, it is logically possible that life precludes the complete determination of its underlying physico-chemical structure. Even if one holds this second view one would probably recommend for biological research no other method than has been pursued in the past decades: attempting to explain as much as possible on the basis of the known physico-chemical laws, and describing the behavior of organisms carefully and without theoretical prejudices.

The first of these two views is more common among modern biologists than the second; but the experience available at present is certainly not sufficient to decide between the two views. The preference that is given by many biologists to the first view may be due again to the Cartesian partition, which has penetrated so deeply into the human mind during the past centuries. Since the "res cogitans" was confined to men, to the "I," the animals could have no soul, they belonged exclusively to the "res extensa." Therefore, the animals can be understood, so it is argued, on the same terms as matter in general, and the laws of physics and chemistry together with the concept of history must be sufficient to explain their behavior. It is only when the "res cogitans" is brought in that a new situation arises which will require entirely new concepts. But the Cartesian partition is a dangerous oversimplification and it is therefore quite possible that the second view is the correct one.

Quite apart from this question, which cannot be settled yet, we are obviously still very far from such a coherent and closed set of concepts for the description of biological phenomena. The degree of complication in biology is so discouraging that one can at present not imagine any set of concepts in which the con-

nections could be so sharply defined that a mathematical representation could become possible.

If we go beyond biology and include psychology in the discussion, then there can scarcely be any doubt but that the concepts of physics, chemistry, and evolution together will not be sufficient to describe the facts. On this point the existence of quantum theory has changed our attitude from what was believed in the nineteenth century. During that period some scientists were inclined to think that the psychological phenomena could ultimately be explained on the basis of physics and chemistry of the brain. From the quantum-theoretical point of view there is no reason for such an assumption. We would, in spite of the fact that the physical events in the brain belong to the psychic phenomena, not expect that these could be sufficient to explain them. We would never doubt that the brain acts as a physico-chemical mechanism if treated as such; but for an understanding of psychic phenomena we would start from the fact that the human mind enters as object and subject into the scientific process of psychology.

Looking back to the different sets of concepts that have been formed in the past or may possibly be formed in the future in the attempt to find our way through the world by means of science, we see that they appear to be ordered by the increasing part played by the subjective element in the set. Classical physics can be considered as that idealization in which we speak about the world as entirely separated from ourselves. The first three sets correspond to this idealization. Only the first set complies entirely with the "a priori" in the philosophy of Kant. In the fourth set, that of quantum theory, man as the subject of science is brought in through the questions which are put to nature in the a priori terms of human science. Quantum theory does not

allow a completely objective description of nature. In biology it may be important for a complete understanding that the questions are asked by the species man which itself belongs to the genus of living organisms, in other words, that we already know what life is even before we have defined it scientifically. But one should perhaps not enter into speculations about the possible structure of sets of concepts that have not yet been formed.

When one compares this order with older classifications that belong to earlier stages of natural science one sees that one has now divided the world not into different groups of objects but into different groups of connections. In an earlier period of science one distinguished, for instance, as different groups minerals, plants, animals, men. These objects were taken according to their group as of different natures, made of different materials, and determined in their behavior by different forces. Now we know that it is always the same matter, the same various chemical compounds that may belong to any object, to minerals as well as animals or plants; also the forces that act between the different parts of matter are ultimately the same in every kind of object. What can be distinguished is the kind of connection which is primarily important in a certain phenomenon. For instance, when we speak about the action of chemical forces we mean a kind of connection which is more complicated or in any case different from that expressed in Newtonian mechanics. The world thus appears as a complicated tissue of events, in which connections of different kinds alternate or overlap or combine and thereby determine the texture of the whole.

When we represent a group of connections by a closed and coherent set of concepts, axioms, definitions and laws which in turn is represented by a mathematical scheme we have in fact isolated and idealized this group of connections with the purpose

of clarification. But even if complete clarity has been achieved in this way, it is not known how accurately the set of concepts describes reality.

These idealizations may be called a part of the human language that has been formed from the interplay between the world and ourselves, a human response to the challenge of nature. In this respect they may be compared to the different styles of art, say of architecture or music. A style of art can also be defined by a set of formal rules which are applied to the material of this special art. These rules can perhaps not be represented in a strict sense by a set of mathematical concepts and equations, but their fundamental elements are very closely related to the essential elements of mathematics. Equality and inequality, repetition and symmetry, certain group structures play the fundamental role both in art and in mathematics. Usually the work of several generations is needed to develop that formal system which later is called the style of the art, from its simple beginning to the wealth of elaborate forms which characterize its completion. The interest of the artist is concentrated on this process of crystallization, where the material of the art takes, through his action, the various forms that are initiated by the first formal concepts of this style. After the completion the interest must fade again, because the word "interest" means: to be with something, to take part in a process of life, but this process has then come to an end. Here again the question of how far the formal rules of the style represent that reality of life which is meant by the art cannot be decided from the formal rules. Art is always an idealization; the ideal is different from reality—at least from the reality of the shadows, as Plato would have put it—but idealization is necessary for understanding.

This comparison between the different sets of concepts in

natural science with different styles of art may seem very far from the truth to those who consider the different styles of art as rather arbitrary products of the human mind. They would argue that in natural science these different sets of concepts represent objective reality, have been taught to us by nature, are therefore by no means arbitrary, and are a necessary consequence of our gradually increasing experimental knowledge of nature. About these points most scientists would agree; but are the different styles of art an arbitrary product of the human mind? Here again we must not be misled by the Cartesian partition. The style arises out of the interplay between the world and ourselves, or more specifically between the spirit of the time and the artist. The spirit of a time is probably a fact as objective as any fact in natural science, and this spirit brings out certain features of the world which are even independent of time, are in this sense eternal. The artist tries by his work to make these features understandable, and in this attempt he is led to the forms of the style in which he works.

Therefore, the two processes, that of science and that of art, are not very different. Both science and art form in the course of the centuries a human language by which we can speak about the more remote parts of reality, and the coherent sets of concepts as well as the different styles of art are different words or groups of words in this language.